Marine microbiology
Lecture 9

Bacterial growth and measurement of growth

Introduction

Synopsis:

This lecture concerns:

- the growth of bacteria
- the process of population growth
- how growth is measured
- applications of understanding bacterial growth

Outcomes:

At the end of this lecture you will:

- be able to differentiate between cell growth and population growth
- understand the significance and the process of population growth
- know how bacterial population growth is measured
- know how an understanding of bacterial growth may be applied

Most bacteria grow by binary fission二分裂

(a) A young cell at early phase of cycle

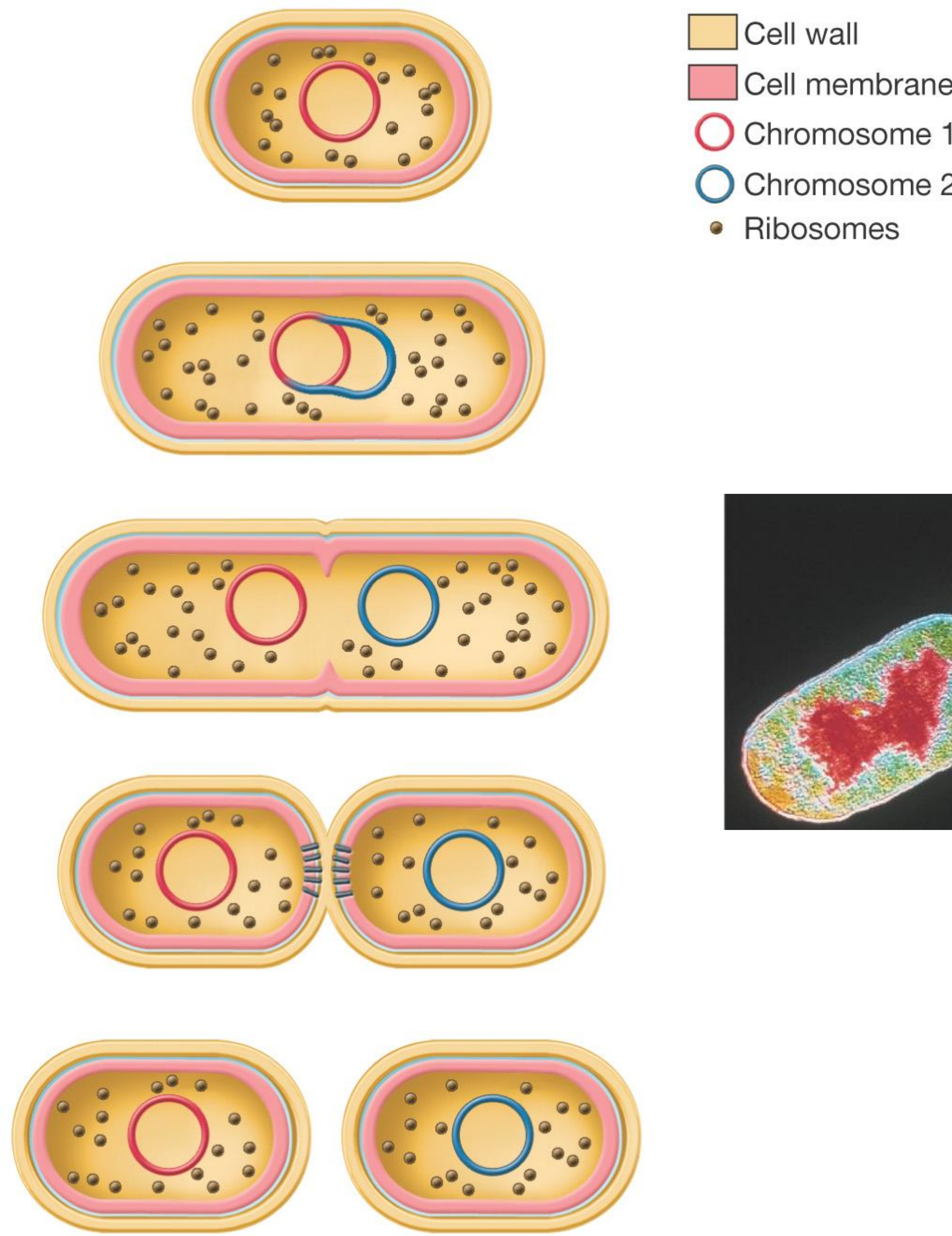
(b) A parent cell prepares for division by enlarging its cell wall, cell membrane, and overall volume.

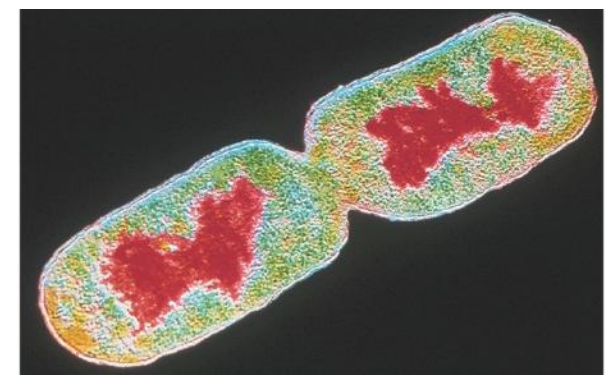
The chromosome then replicates.

(c) The septum begins to grow inward as the chromosomes move toward opposite ends of the cell. Other cytoplasmic components are distributed to the two developing cells.

(d) The septum is synthesized completely through the cell center, and the cell membrane patches itself so that there are two separate cell chambers.

(e) At this point, the daughter cells are divided. Some species separate completely as shown here, while others remain attached, forming chains, doublets, or other cellular arrangements.





Bacterial binary fission

- As a result of binary fission, populations increase in number
- This increase is **logarithmic:** the number doubles at each generation

  i.e. 1 – 2 – 4 – 8 – 16 – 32 – 64 – 128 – 256 – 512 – 1024 –
- The generation time (g) is the time for a cell to produce 2 cells or the time for a bacterial population to double in number
- For ***E. coli*** g = 15-20 mins (under optimal conditions)
- for ***Mycobacterium tuberculosis***（结核杆菌） g = ~ 12 hours.

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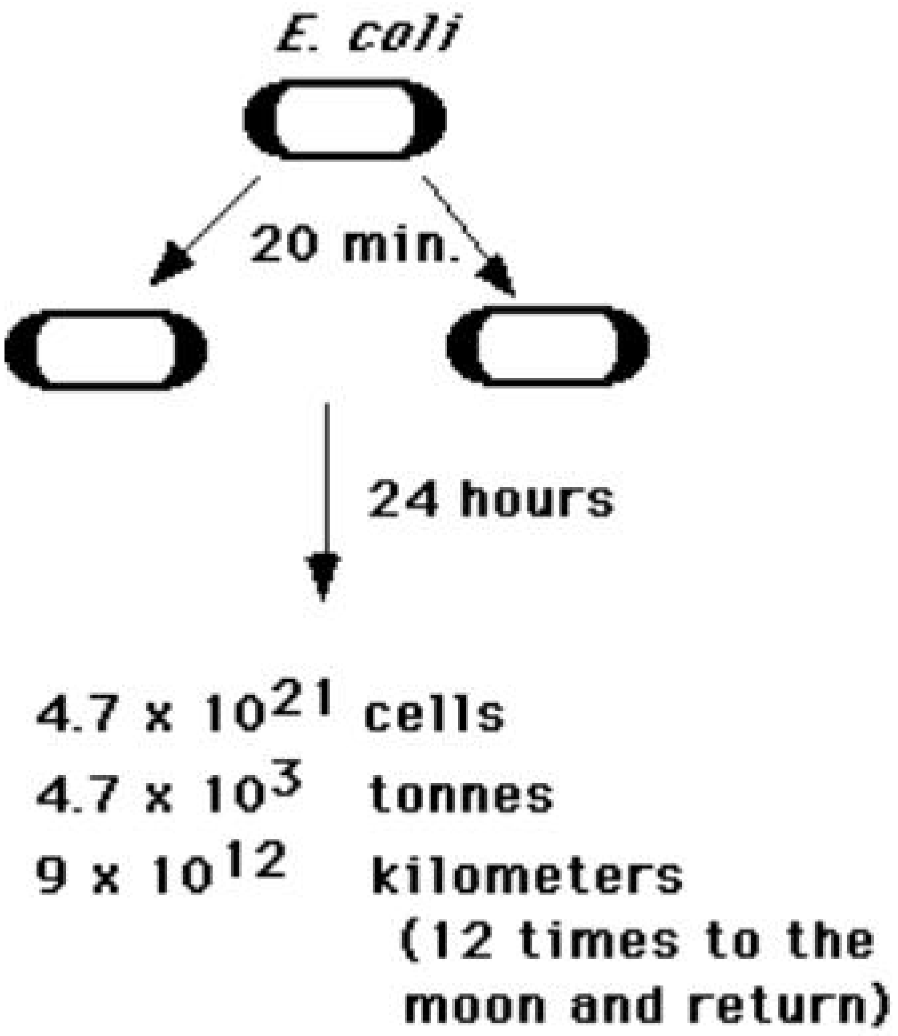
The significance of bacterial growth

If exponential（指数的）growth continued, microbes would take over the planet!

In reality, as the number of bacteria increases in any one environment:

- nutrients are used up
- metabolic wastes accumulate
- living space may become limited
- aerobes may suffer from oxygen depletion

Thus bacterial growth stops.



The significance of bacterial growth

It has human health implications

- bacterial meningitis（脑膜炎）, cholera（霍乱）, food poisoning from staphylococci（葡萄球菌）or salmonella（沙门氏菌）, plague（瘟疫）, tetanus（破伤风）etc.

It has animal health implications

- anthrax（炭疽）in ruminants（反刍动物）, mastitis（乳腺炎）in dairy herds（乳畜群）etc.

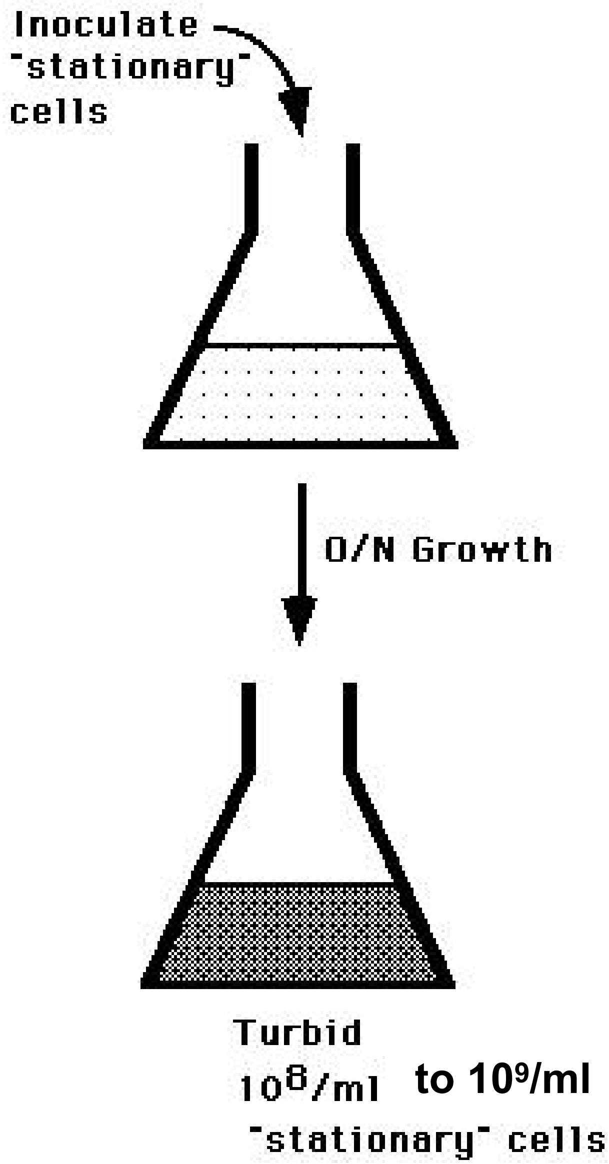
It has environmental impacts

- **eutrophication富营养化** (enrichment of aquatic environment with nutrients e.g. from sewage污水 pollution)
- blooms（水华）of toxic cyanobacteria（蓝藻细菌）
- acid sulphate soils
- composting（肥料）and sewage treatment

It provides for industrial applications

- producing products (e. g. cheese, antibiotics, insulin胰岛素)
- industrial processes (e. g. leaching metals金属浸出 from low grade ores矿)

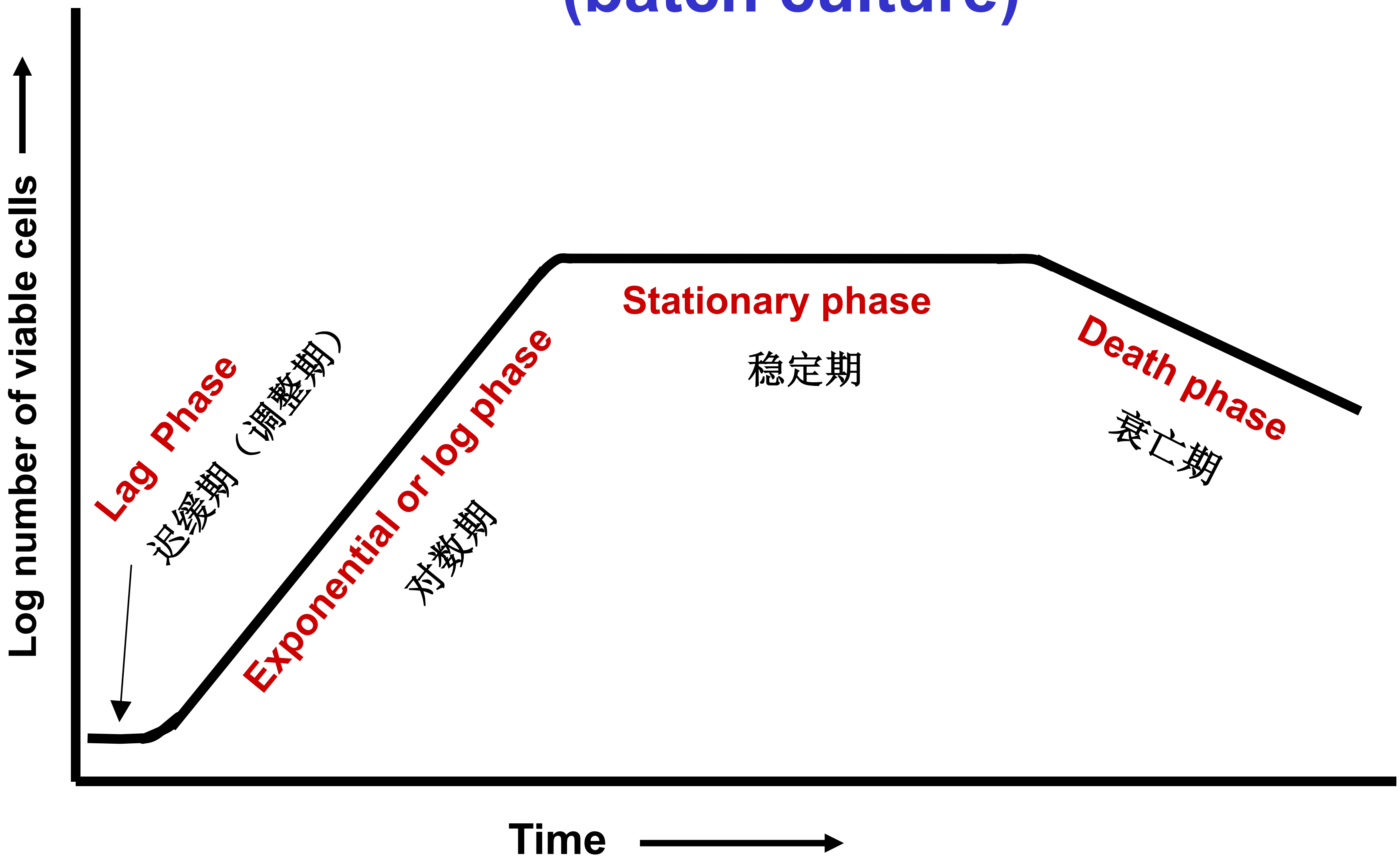
The population growth cycle



Unravelling the overnight culture

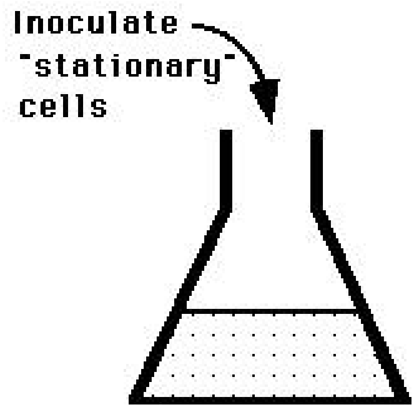
- A few non-dividing ("stationary") cells are inoculated into fresh medium and begin to grow and multiply（繁殖）
- After overnight growth there are about **10^9/ml** stationary (viable but non-dividing) cells in the medium
- What occurs between the **initiation** of growth and **cessation** (停止) of growth?

The bacterial growth curve in a closed system (batch culture)



Growth curve of a typical bacterium grown in batch culture illustrating the four phases; lag phase, exponential growth phase, stationary phase and decline or death phase

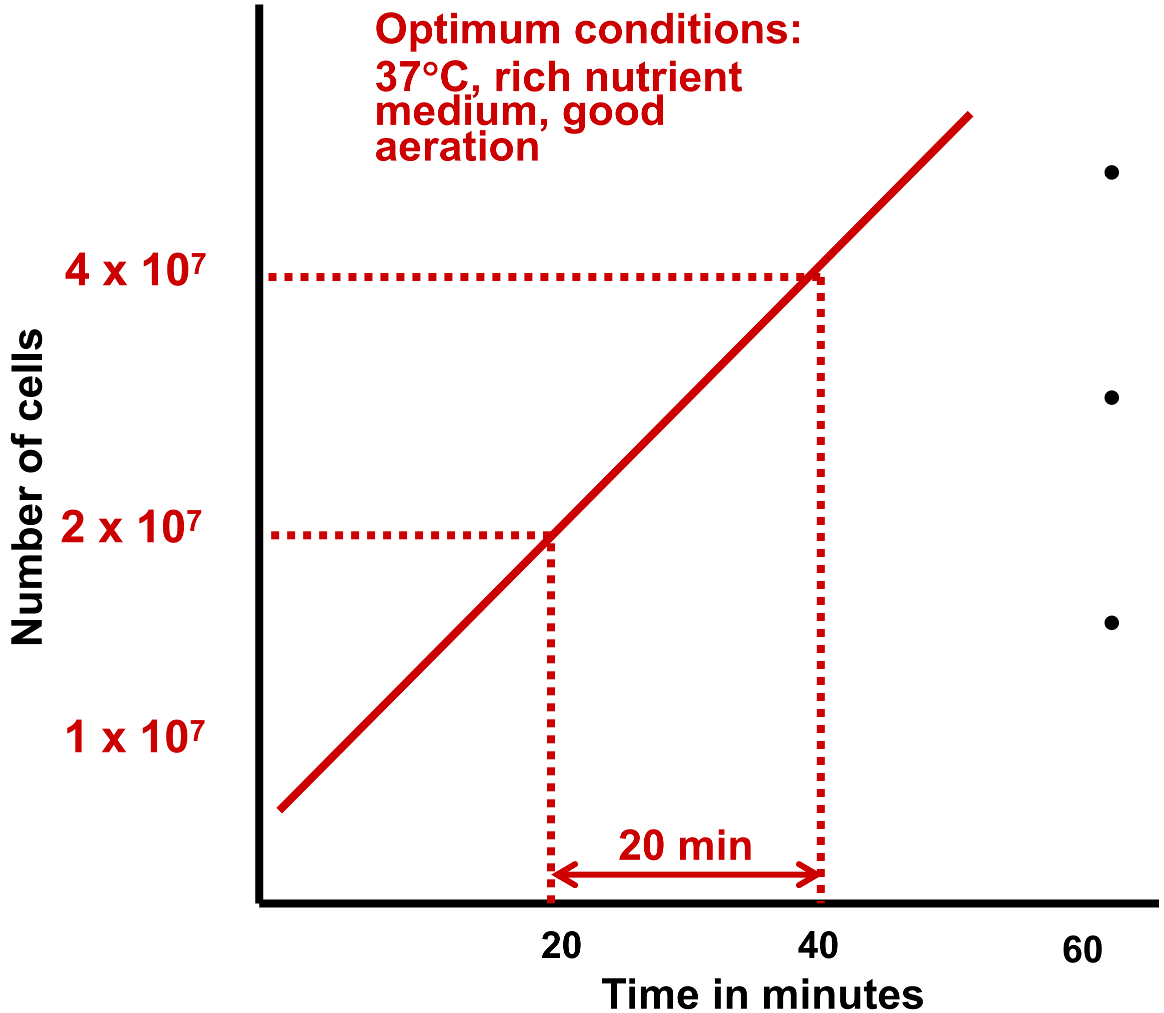
The population growth cycle



1. **Resuming（恢复）growth – The Lag Phase**

- Cells have to **adapt** to new physiological conditions
- Synthesis of cellular constituents 细胞组分 (especially ribosomes and enzymes) begins
- Energy (ATP) becomes available for growth
- Cell **volume increases** but no cell division so the growth rate (k) and generation time (g) are zero
- Rates of physiological processes are initially **unbalanced**
  - different genes are upregulated at different times
- **DNA replication** and **cell division** initiated
  - cell division, and hence population growth, begins

The population growth cycle



Generation time (g) in an exponentially growing bacterial population. Log scale on y axis.

2. Exponential (or Log) Phase

- Growth is **balanced** - the relative synthesis rates are constant (DNA, ribosomes, enzymes, cell walls etc.)
- Population growth rate (k) and generation (or doubling) time (g) are **constant**
- A graph of log [No.] vs Time produces a **straight line**, with slope = k

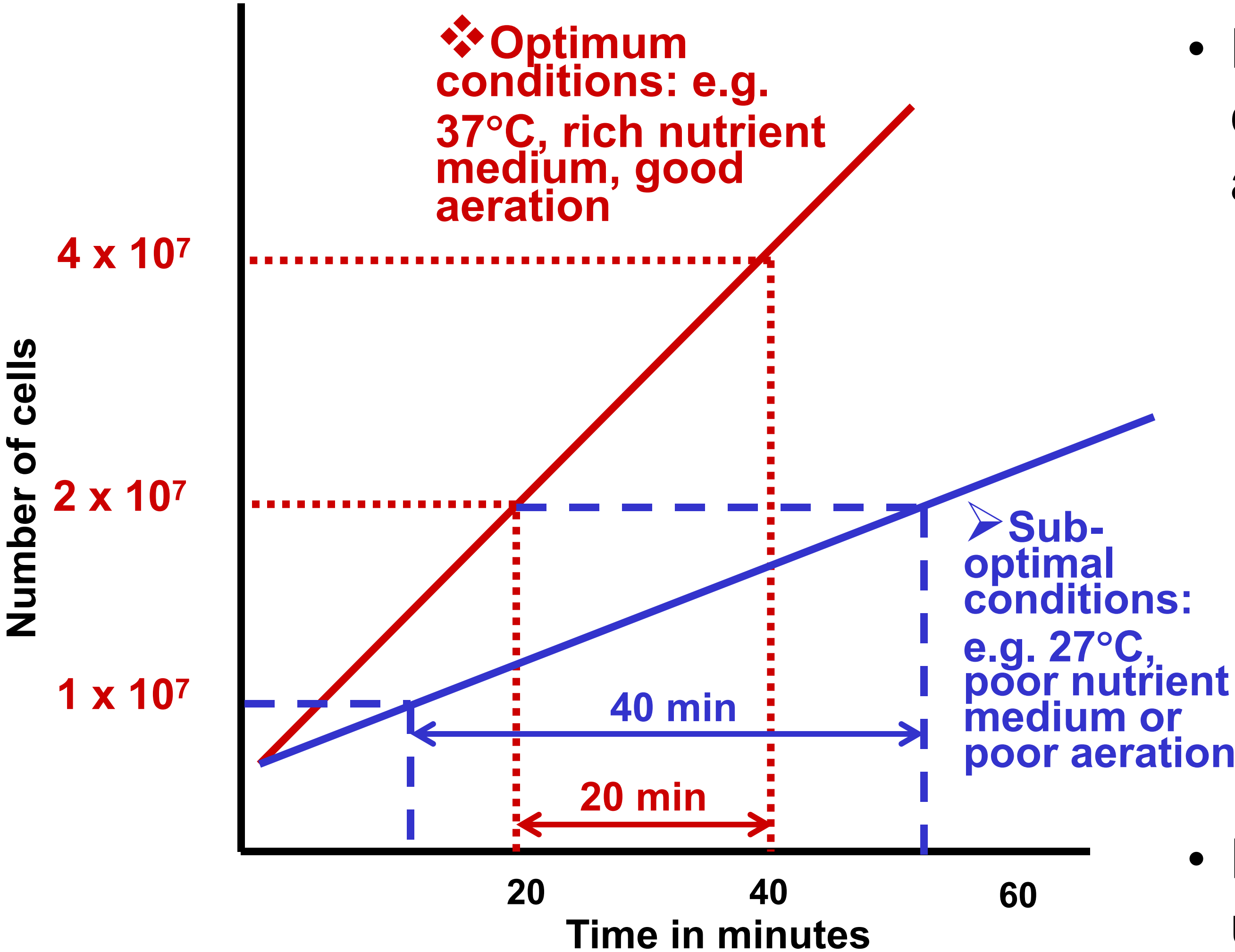
$k = 1/g$ (generations/hour)

e.g. if g = 20 mins (0.33 hr)

k = 1/0.33 = 3.03 gens/hr

The population growth cycle

2. Exponential (or Log) Phase



Generation time (g) of *E. coli* under different conditions.

- Rate of growth depends on conditions such as temperature and nutrients
  - under different (stable) conditions exponential growth occurs but at a different rate
    - ❖ k = 3.03 gens/hr
    - ➢ k = 1.49 gens/hr
- Exponential growth continues until some factor becomes limiting
- Limiting factors drive the culture into **stationary** phase

Table 7.3 Examples of Generation Times[a]

Microorganism	*Incubation Temperature (°C)*	*Generation Time (Hours)*
Bacteria		
Escherichia coli	40	0.35
Bacillus subtilis	40	0.43
Staphylococcus aureus	37	0.47
Pseudomonas aeruginosa	37	0.58
Clostridium botulinum	37	0.58
Mycobacterium tuberculosis	37	≈12
Treponema pallidum	37	33
Protists		
Tetrahymena geleii	24	2.2–4.2
Chlorella pyrenoidosa	25	7.75
Paramecium caudatum	26	10.4
Euglena gracilis	25	10.9
Giardia lamblia	37	18
Ceratium tripos	20	82.8
Fungi		
Saccharomyces cerevisiae	30	2
Monilinia fructicola	25	30

[a] Generation times differ depending on the growth medium and environmental conditions used.

The population growth cycle

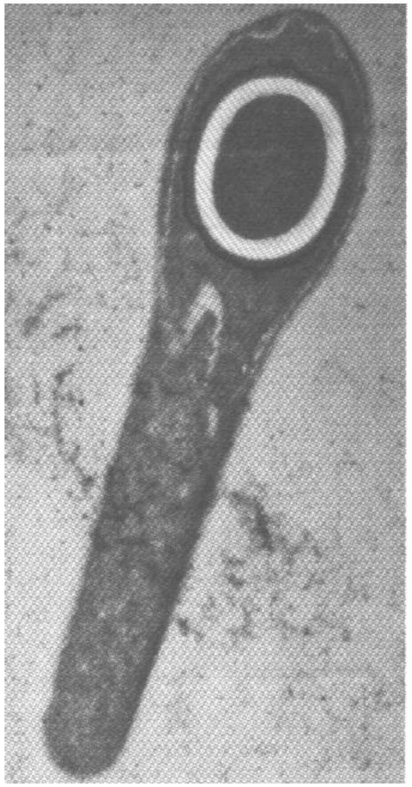
3. The Stationary Phase of growth

Total viable cell numbers remain **constant (*k*=0)**

- Cells are not dividing but remain metabolically active because nutrients have been depleted

or

- a metabolic waste/toxin may have built up and cell death or inactivation balances cell division



Properties of cultures in stationary phase

- Cells tend to be **small**
- Cells have a **different composition** of cellular constituents compared to growing cells
- Growth again **unbalanced**
- Available energy is used for **maintenance** (cell survival mechanisms e.g. endospore formation may be initiated if species has that ability)

The population growth cycle

3. The Stationary Phase of growth

What factors limit growth in batch culture?

- Bacteria consume resources and produce wastes

- Nutrient limitation or inhibition by toxic waste products occurs
  - e.g. Streptococcal cultures can produce sufficient lactic acid (乳酸) to cause acid inhibition
  - e.g. aerobic microbes may be limited by O_2 availability

- Growth resumption requires nutrient addition and/or toxin removal

The population growth cycle

4. The Death Phase

<u>The "classical view":</u>

- ➢ Cells begin to die as nutrient deprivation or toxic waste build-up（积累）cause irreparable（不能修复的）damage to the cells
  - *k* becomes negative

- ➢ Population death rate is usually logarithmic
  - a constant proportion dies per hour

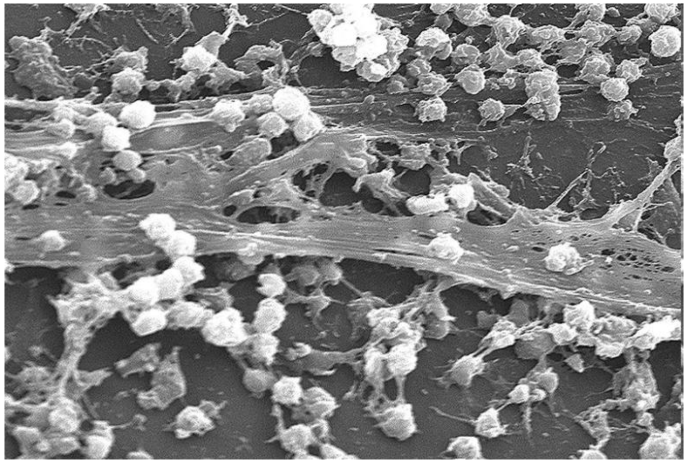
- ➢ Death rate may slow down later (may not be logarithmic)
  - resistant cells, if present, constitute an increasing proportion of remaining viable cells

<u>Other hypotheses are currently being debated:</u>

➢does a fraction of the population of cells die due to activation of programmed cell death（程序性细胞死亡）genes? The dead cells then release nutrients which support the growth of other cells.

What about bacterial population growth in the natural world?

- As in culture, cycles of cell growth occur in nature
- Bacteria in soils, waters, surfaces, intestinal tracts (肠道) are subject（受影响）to stimulants（兴奋剂）and inhibitors（抑制剂）of growth
- Natural systems are very complex - variable in physical, chemical and biological factors affecting populations
  - Many microbes form **biofilms:** aggregations（聚集）of microbes in complex communities, growing on surfaces and held together by extracellular polymers e.g. polysaccharides
- Growth cycles will fluctuate（波动）to reflect environmental variability and complexity



S. aureus biofilm on a catheter导尿管

How is bacterial growth measured?

Bacterial growth can be measured directly or indirectly

A. Direct

Measure cell biomass

Wet weight (pellet) **or Dry weight** (dry pellet in oven烘箱) **of cells**

- **Insensitive** (> 10^9 cells/ml required), **very slow**
- Used for studies on fungi and filamentous bacteria（丝状细菌）

Measure cell number

1. **Direct (or total) cell counts**
   - e. g. microscope counts
   - **not very sensitive** (**> 10^7** cells/ml required)
   - **rapid** method (completed in ~30mins)
2. **Viable cell counts**
   - e.g. dilution plating, concentration (filtration)
   - **sensitive** (can detect < 10 cells/ml)
   - **slow**, depends on bacterial growth rate e.g.<24hr

How is bacterial growth measured?

A. DIRECT

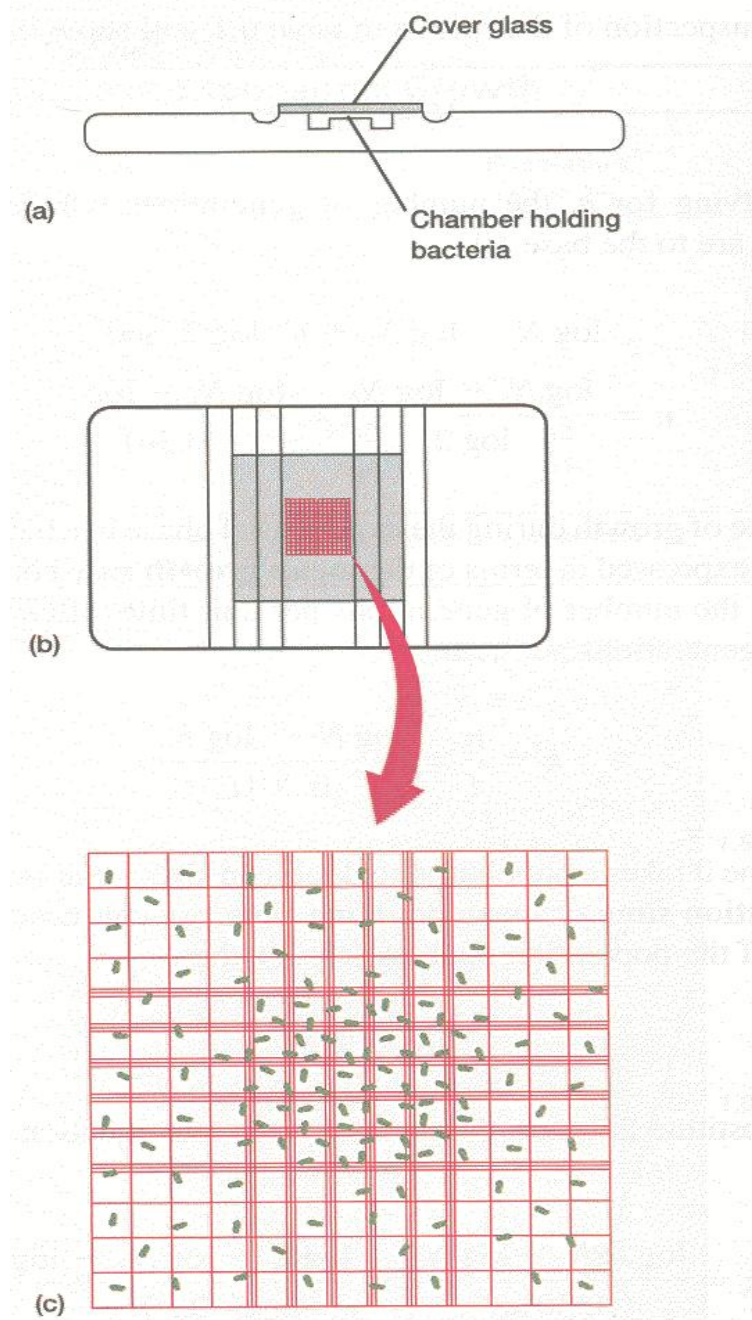
1. Direct (total) cell counts

Uses a counting chamber

(a) Side view of chamber showing coverglass and space beneath that holds bacterial suspension

(b) Top view of the chamber. The grid is located in the center of the slide

(c) An enlarged view of the grid. The bacteria in several squares are counted. The number in the square is used to calculate the concentration of cells in the original sample



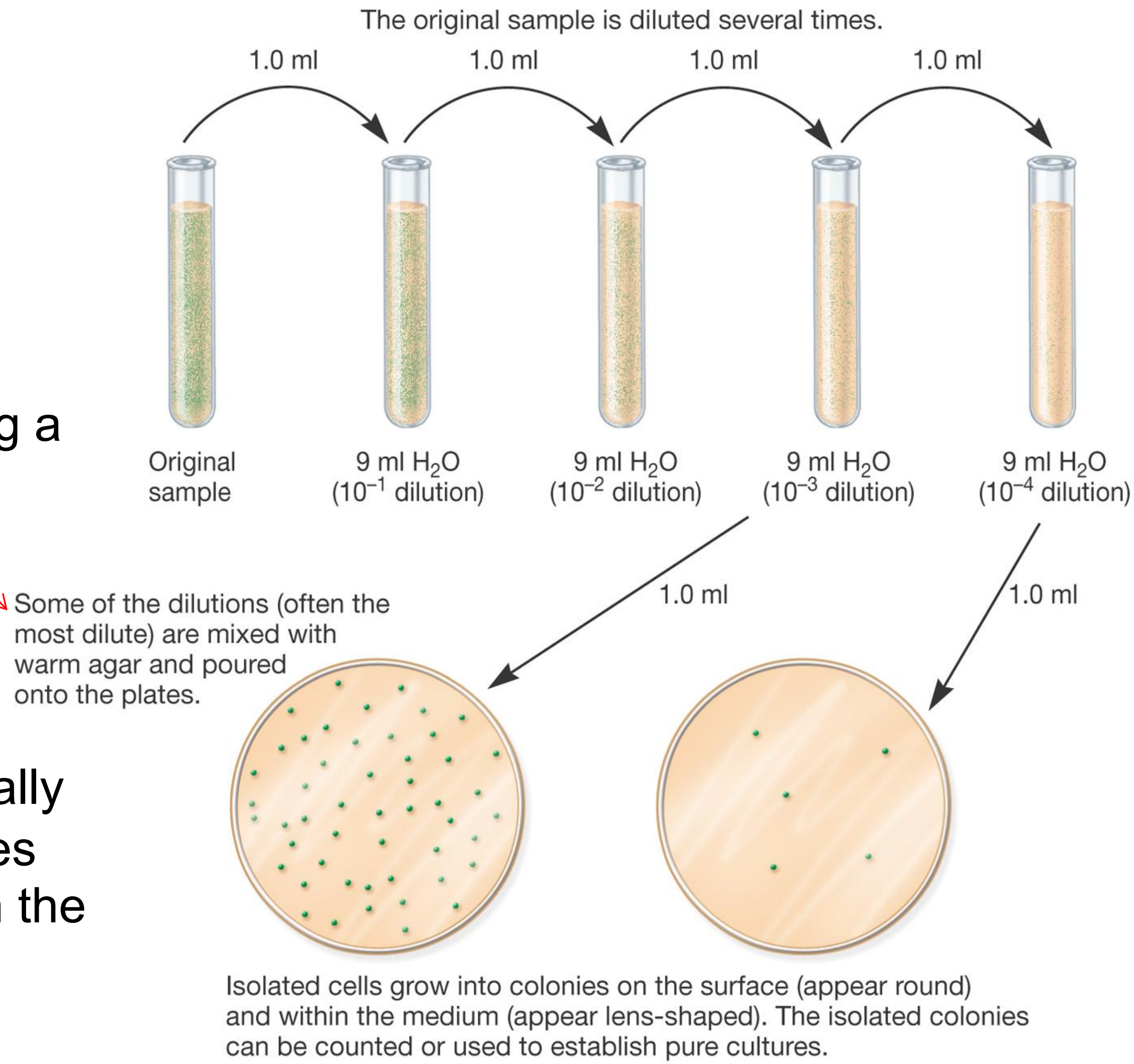
How is bacterial growth measured?

A. DIRECT

2. Viable cell counts

(a) Serial dilution（连续梯度稀释）and plating

Usually 0.1ml aliquots are taken and spread onto the surface of agar plates, using a sterile glass or plastic spreader



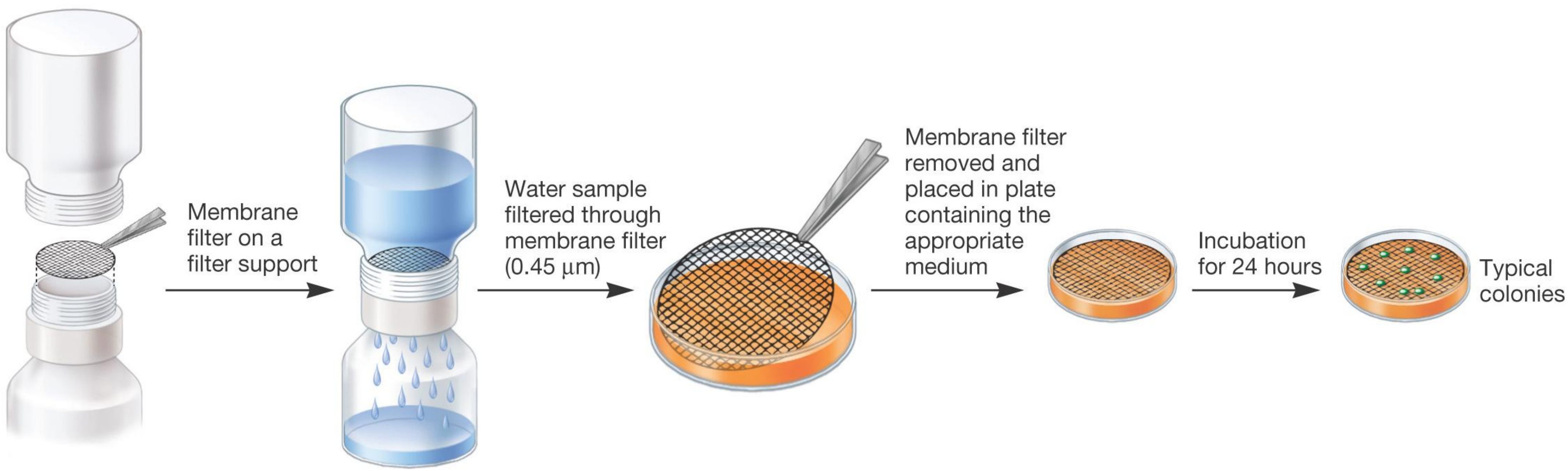
For statistical reasons ideally plates with 30–300 colonies are counted, if available in the series

How is bacterial growth measured?

A. DIRECT

2. Viable cell counts

(b) Concentration（浓缩）/filtration（过滤）



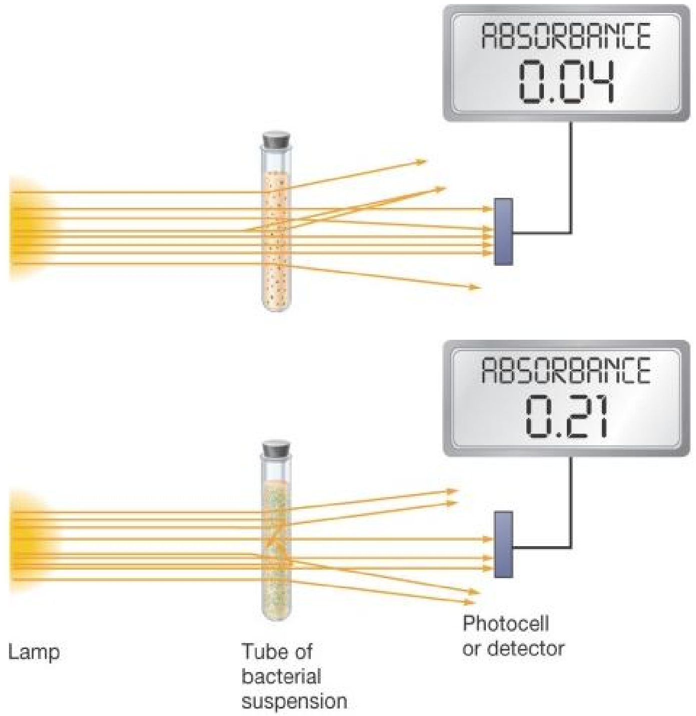
The Membrane Filtration Procedure. Membranes with different pore sizes (e.g. 0.45μm) are used to trap different microorganisms. Incubation times for membranes also vary with the medium and microorganism

How is bacterial growth measured?

B. Indirect growth measures

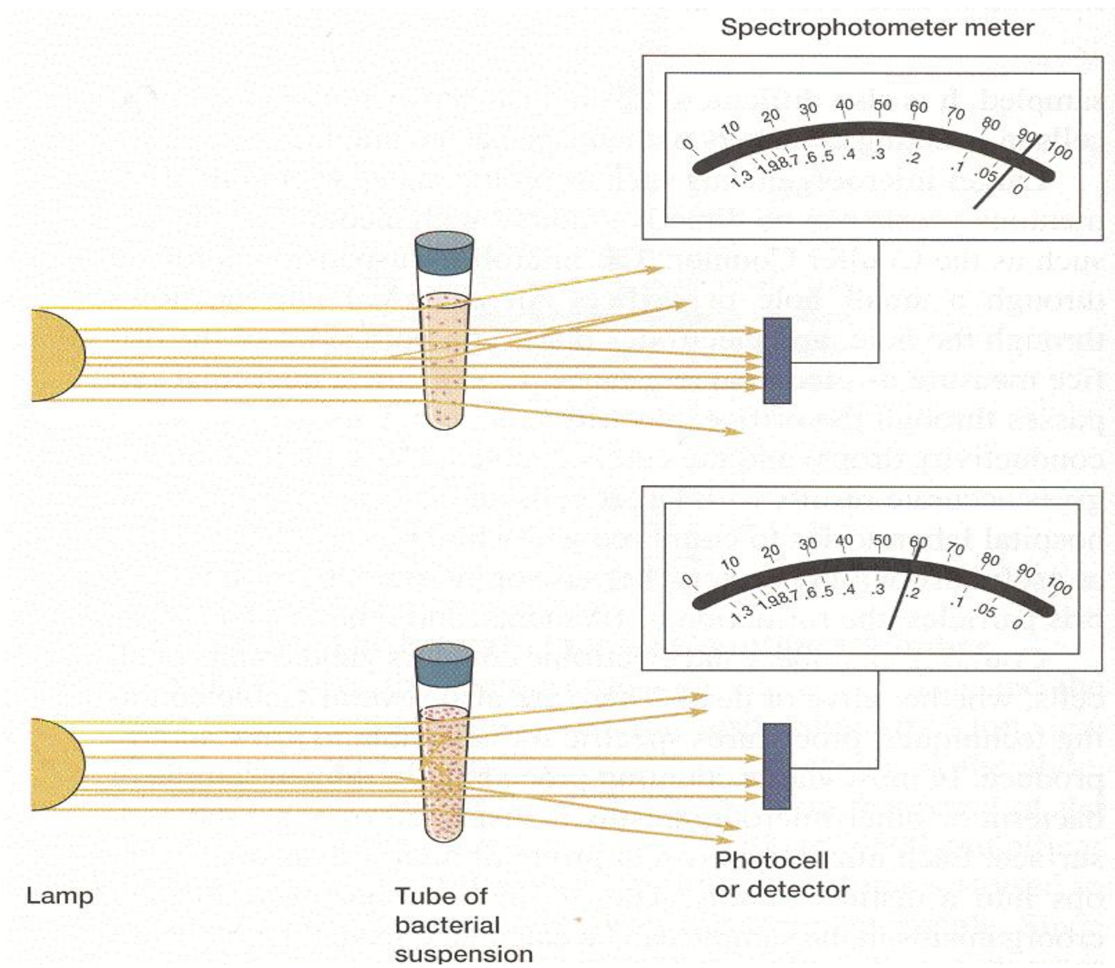
1. Turbidity（浊度）

- bacterial cells affect the passage of light through a solution
- measured by light absorbance（吸光度）OR its inverse, light transmittance (透射率)
- a useful indirect measure
- **very rapid** method and often used
- **inaccurate** when cell densities are high



How is bacterial growth measured?

Measuring turbidity



- The spectrophotometer（分光光度计）has 2 scales.
  - The bottom scale displays **absorbance** and the top scale, **percent transmittance.**
- Absorbance increases as % transmittance decreases
  - absorbance = 1 when % transmittance = 0%
  - absorbance = 0 when % transmittance = 100%

Turbidity and microbial mass measurement Determination of microbial number by measurement of light absorption. As the population and turbidity increase, more light is scattered（分散的）and so the absorbance reading increases.

How is bacterial growth measured?

B. Indirect growth measures

2. Cell mass – quantity of cell constituent（组分）

- if amount of a substance in each cell is constant, total quantity of the cell constituent is directly related to total microbial mass
- e.g. measure in a sample of washed cells from a known volume:
  - total protein
  - total nitrogen
  - total ATP

Applications of knowledge of bacterial growth

- **For many purposes it is desirable to maintain cells in exponential growth**
  - **Useful in genetic and physiological experiments:**
    - cells in identical growth state
    - relative synthetic rates constant for all cell constituents
    - thus reproducibility in repeated experiments
  - **Useful in industrial fermentations, to make:**
    - **Primary metabolites**（初级代谢产物）: produced during active growth

      e. g. organic acids, enzymes, amino acids
- **For other purposes, stationary phase is needed**
  - **Secondary metabolites** （次级代谢产物）: products of metabolism synthesized after growth has been completed

    e. g. antibiotics (Bacitracin杆菌肽 from ***Bacillus*** sp.枯草芽孢杆菌) produced in stationary and death phases

Applications of knowledge of bacterial growth

- A **chemostat（恒化器）** enables cells to be kept in the exponential phase, at a specific growth rate, by providing a constant supply of one essential nutrient at a limiting concentration = **a continuous culture system**

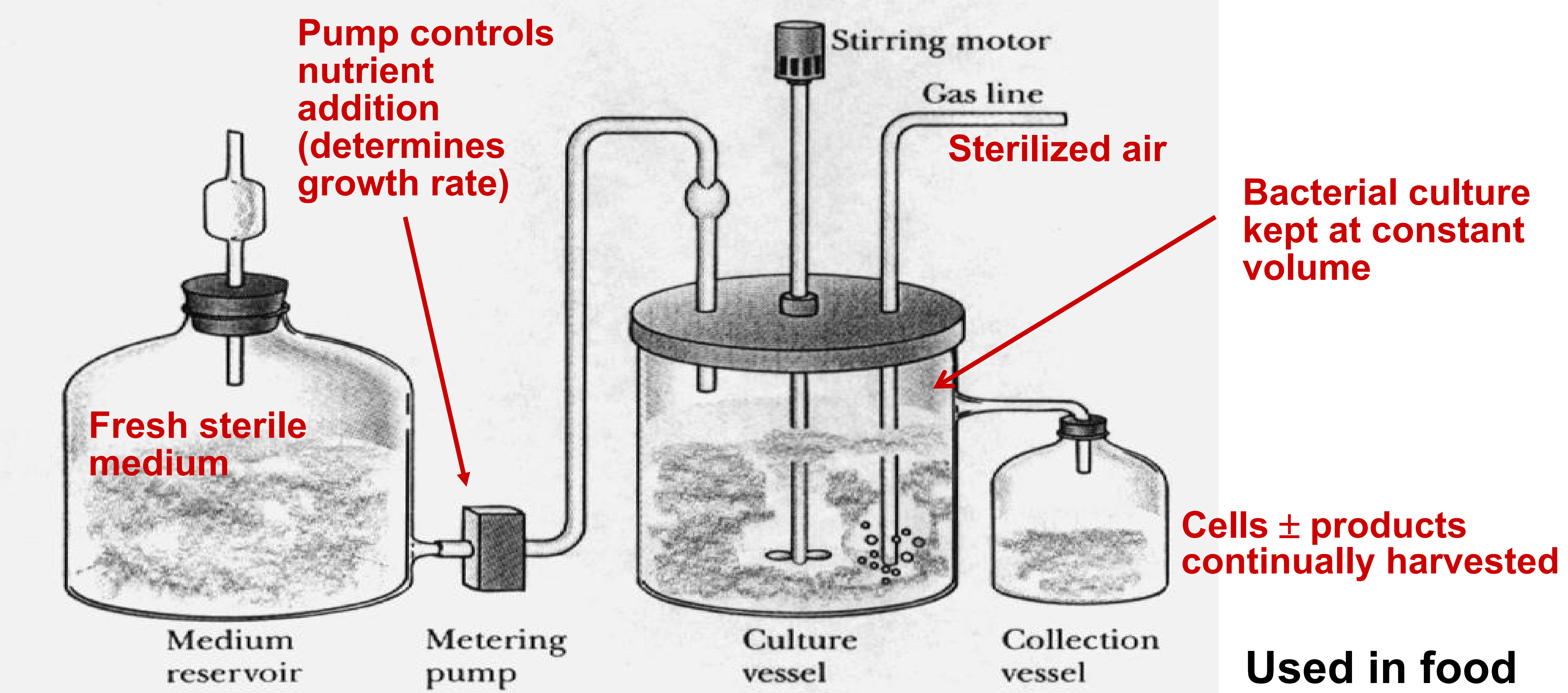


Figure **6.14** Model of a chemostat apparatus for the continuous growth of a bacterial culture.

Used in food and industrial microbiology

Summary

- Most bacterial cells reproduce by binary fission
- Bacterial population growth has implications ranging from human and environmental health to industrial applications
- Bacterial population growth involves 4 phases; lag, exponential, stationary and death phases
- Bacterial growth in nature is complicated by the complexity of the natural world; biofilms are common
- Bacterial biomass and numbers are measured by a variety of direct and indirect methods
- Chemostats, which allow maintenance of cultures in the exponential growth phase, have research and industrial application